

Intro to Natural Language Processing for Clinical Text

Matthew Engelhard

Lecture Outline

Part 1: *why* work with clinical text?

- Structured (tabular) versus unstructured (notes, etc) EHR data
- What's in notes and other unstructured data, and why is it important?

Part 2: *how* do we work with unstructured data?

- Extracting clinical text
- Converting to a format we can use
- Simple models for text

Part 1: *why*

...do we care about clinical text and other unstructured data?

Structured Data

Definition: data organized into specific, queryable fields as part of an overarching model, with each field having a defined purpose.

- typically stored in tabular (i.e. table, spreadsheet) format
- numeric variables, categorical variables and/or short free text

Examples: demographics, diagnoses, procedures, labs, meds, vitals

Example from Duke's Clinical Research Datamart...

PCOR_RSH_DIAGNOSIS_VW EXAMPLE

DIAGNOSISID	PATID	ENCOUNTERID	ENC_TYPE	ADMIT_DATE	PROVIDERID	DX	DX_TYPE	DX_SOURCE	DX_ORIGIN	PDX	DX_POA	RAW_DX	RAW_DX_TYPE	RAW_DX_SOURCE	RAW_PDX
36510180	123456	57896636	AV	2014/02/21 00:00:00	1745326	530.9	9	FI	BI	OT	NI	530.9	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36510407	223456	57897312	AV	2014/05/07 00:00:00	73411	530.81	9	FI	BI	OT	NI	530.81	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36510405	323456	57894371	ED	2014/06/13 00:00:00	123901	553.3	9	FI	BI	OT	NI	553.3	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY
26510406	423456	57843289	AV	2015/01/21 00:00:00	1719263	530.3	9	FI	BI	OT	NI	530.3	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY
36510721	523456	57898224	AV	2015/03/04 00:00:00	1719263	787.2	9	FI	BI	OT	NI	787.2	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36510720	612345	57899120	ED	2015/07/30 00:00:00	156342	454.8	9	FI	BI	OT	NI	454.8	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36511049	654321	57892268	EI	2016/12/03 00:00:00	165493	V12.72	9	FI	BI	OT	NI	V12.72	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY
36508682	543216	57893648	IP	2016/08/05 00:00:00	178238	211.3	9	FI	BI	OT	NI	211.3	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY

COLUMN	DEFINITION
DIAGNOSISID:	An arbitrary identifier for each record. Not currently useful for linkage across tables.
ENCOUNTERID:	The ENCOUNTERID (encounter ID) is an arbitrary identifier for a specific encounter that can be used to link information across tables in the model.
PATID:	The PATID (patient identifier) is an arbitrary, person-level identifier used to link information across tables within the data model. There is a table within the CRDM that allows an investigator with an appropriate IRB protocol to link PATID to MRN.
ENC_TYPE:	Encounter type indicates the setting in which the encounter took place (the same data are provided in the encounter table: AV=Ambulatory Visit, ED=Emergency Department, EI=Emergency Department Admit to Inpatient Hospital Stay, IP=Inpatient Hospital Stay, OS=Observation Stay, NI=No information, UN=Unknown, OT=Other
ADMIT_DATE:	The date on which the encounter began (also included in the encounter table).
PROVIDERID:	Unique identifier for a provider ? can be used to link an encounter to the provider table.
DX:	The diagnosis code; uses either ICD-9 or ICD-10
DX_TYPE:	09=ICD-9-CM, 10=ICD-10-CM, 11=ICD-11-CM, SM=SNOMED CT, NI=No information,

Unstructured Data

Definition: data that is NOT structured data

- in other words, everything else
- typically stored in a modality-specific format

Examples:

- clinical notes (also available through CRDM)
- imaging (e.g. radiology, ophthalmology, ultrasound, pathology)
- physiologic time series (e.g. electrophysiologic studies, digital health)

What information is in notes

...and why is it important?

A patient presents with shortness of breath...

- Presents to urgent care and is transferred to the ED
- Evaluated in the ED
 - History of present illness
 - Review of medical history (including meds and allergies) + social and family hx
 - Physical exam and review of systems
 - Lab results: BMP and CBC
 - Imaging & other studies: CT and EKG
- Admitted to the hospital

What will we find in the structured data?

- Encounter details (ED admission)
 - Labs (BMP, CBC)
 - Vital signs
 - Procedure codes (CT)
 - Diagnosis codes (PE)
-
- ...what *won't* we find?

The H&P

written by a UNC student and publicly available:

<https://www.med.unc.edu/medclerk/education/grading/history-and-physical-examination-h-p-examples/>

FHIR systems **SHALL** support, at *minimum*, these eight “Common Clinical Notes”:

1. Consultation Note (11488-4)
2. Discharge Summary (18842-5)
3. History & Physical Note (34117-2)
4. Procedures Note (28570-0)
5. Progress Note (11506-3)
6. Imaging Narrative (18748-4)
7. Laboratory Report Narrative (11502-2)
8. Pathology Report Narrative (11526-1)

Chief Complaint:

Shortness of breath.

History of the Present Illness:

Mr. ■ is a previously healthy 56-year-old gentleman who presents with a four day history of shortness of breath, hemoptysis, and right-sided chest pain. He works as a truck driver, and the symptoms began four days prior to admission, while he was in Jackson, MS. He drove from Jackson to Abilene, TX, the day after the symptoms began, where worsening of his dyspnea and pain prompted him to go to the emergency room. There, he was diagnosed with pneumonia and placed on Levaquin 500 mg daily and Benzonatate 200 mg TID, which he has been taking for two days with only slight improvement. He then drove from Abilene back to Greensboro, where he resides, and continued to experience shortness of breath, right sided chest pain, and hemoptysis. He presented to an urgent care office in town today, and was subsequently transferred to the Moses Cone ER due to the provider’s suspicion of PE.

The right-sided pain is located midway down his ribcage, below the axilla. This pain is sharp, about 7/10 in severity, and worsens with movement and cough. Pressing on the chest does not recreate the pain. He feels that the pain has improved somewhat over the past two days. The hemoptysis has been unchanged since it began; there is not frank blood, but his sputum has been consistently blood-tinged. The blood seems redder at night. The dyspnea has been severe, and it is difficult for him to walk more than across a room. He states that he feels as though there is a “rattling” in his chest. At baseline, he experiences no dyspnea on exertion and has no history of COPD or other respiratory problem. He is a smoker, smoking a little less than a pack a day for thirty-five years. Past history is notable for the fact that he experienced transient left lower leg swelling – from below the knee down – and pain several weeks ago during a cross-country haul. He also notes a four day history of decreased appetite, poor sleep, and subjective fever and chills, with a measured fever of 103 in the hospital in Abilene. He had a bout of pneumonia about two months ago, but has been healthy for the most part and denies any chronic medical conditions. Currently he is fairly comfortable, with morphine helping with the pain. He has no history of a clotting disorder, no cardiac history, and denies any chest trauma or aspiration. He has had no sick contacts.

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Past Medical History:

1. Hernia repair
2. Bilateral thumb surgeries, secondary to two separate injuries sustained while working with machinery

Medications:

No regular medications, over-the-counter medications, or supplements. Has taken two days of the medications prescribed by the ER in Abilene: Levaquin 500 mg daily and Benzonatate 200 mg TID.

Review of Systems:

Constitutional: Denies changes in weight, fatigue, night-sweats.

HEENT: No changes in vision, nasal discharge, headache.

CV: No palpitations, left-sided chest pain/pressure, edema.

Resp: See HPI

GI: No nausea, vomiting, diarrhea, constipation.

GU: No dysuria, increased frequency.

Neuro: No weakness, confusion, numbness, dizziness.

MSK: No weakness, arthralgias, myalgias.

Heme: No easy bruising, easy bleeding.

Skin: No new lesions or rashes.

Endocrine: No polydipsia, polyuria, heat/cold sensitivity.

Social History:

Mr. [REDACTED] is divorced, lives in [REDACTED], and has been a truck driver for many years. He lives with his brother and his brother's wife. He has several children, including a daughter who also lives in the area. His family is very supportive and they get along well. As per HPI, he smokes a little less than a pack per day and has for thirty-five years. He does not drink alcohol or use other drugs. He tries to be physically active, and his favorite type of exercise is line dancing. He has “excellent” insurance through his employer.

Family History:

No family history of lung disease or clotting disorders.

Mother died at 82 of “old age.”

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Assessment & Plan:

Previously healthy 56 year old man presenting with shortness of breath, hemoptysis, and right-sided chest pain, with CT images demonstrating pulmonary emboli and infiltrates.

1. Shortness of breath: The symptoms of dyspnea, hemoptysis, and the right-sided chest pain could all be explained by a pulmonary embolus, and the CT shows evidence that confirms this diagnosis. Management should be to supply supplemental oxygen to maintain adequate oxygen saturation, to control Mr. ■'s chest pain, and to anticoagulate him. Anticoagulation with low molecular weight heparin and Coumadin could be started simultaneously, with both therapies should overlapping for at least five days; we will seek to obtain an INR between 2 and 3. The patient is currently hemodynamically stable, without evidence of right heart strain, so thrombolysis does not seem indicated at this time. With his history of long drives back and forth to California, Mr. ■ is at risk for venous stasis, which is a major source of DVT's; he even presents a history of lower leg pain and swelling. Testing for various causes of hypercoagulability would also be reasonable: these include deficiencies of proteins C and S, antithrombin III, lupus anticoagulant or anticardiolipin antibodies, and factor V mutations, prothrombin mutation, lipoprotein a, and hyperhomocystinemia. Another important factor to consider would be malignancy, especially as Mr. ■ is a long-time smoker. Screening for lung, colon, and prostate cancer would be appropriate.
2. Possible pneumonia: Mr. ■ has a history of being febrile, and was diagnosed with pneumonia in Abilene (obtaining these records will be helpful), but does not currently have an elevated white count. The CT demonstrates an infiltrate, possibly consistent with pneumonia. Sputum cultures and blood cultures would be reasonable to obtain, though since he has been taking an antibiotic for presumed community acquired pneumonia, this may be low-yield. His vitals should be monitored regularly. It is possible that all pulmonary symptoms stem from the pulmonary embolus, but it would be reasonable to continue the antibiotic therapy that he has already started in order to cover the possibility of an infectious component.
3. Anemia: Admission labs show hemoglobin at 11.5, with MCV of 89.5, so this anemia is normocytic. The patient denies any history of anemia when questioned.

Notes are a rich source of info

Research and Applications

Real world evidence in cardiovascular medicine: ensuring data validity in electronic health record-based studies

Tina Hernandez-Boussard,^{1,2,3} Keri L Monda,^{4,5} Blai Coll Crespo,⁴ and Dan Riskin^{1,3,6}

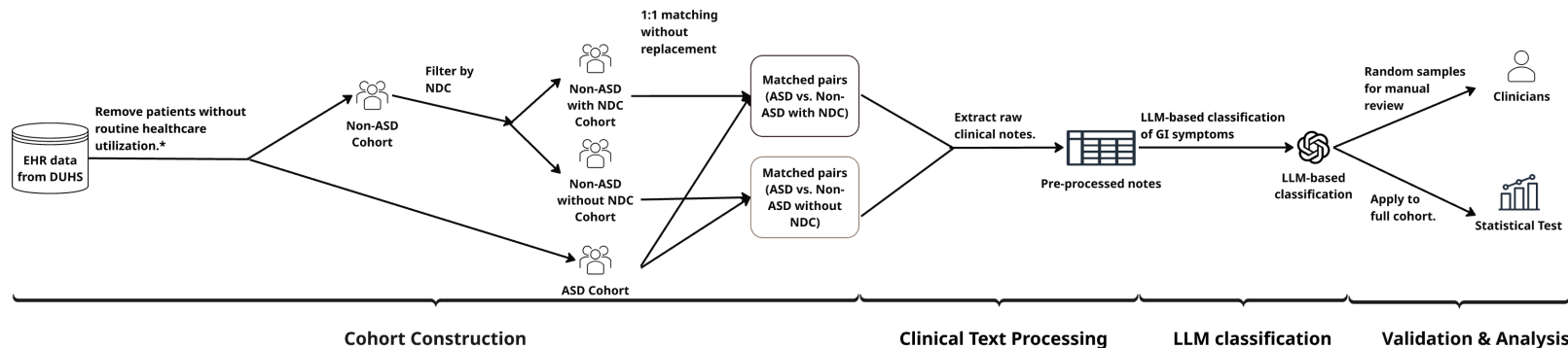
Table 1. Cohort identification of diseases and procedures stratified by EHR-S and EHR-U data^a

Cohort	Occurrence		EHR-S			EHR-U		
	Concept	Patient	Recall (%)	Precision (%)	F1-score (%)	Recall (%)	Precision (%)	F1-score (%)
Hyperlipidemia	2471	837	65.2	99.3	78.7	98.2	99.4	98.8
Hypercholesterolemia	1899	478	55.1	98.0	70.5	90.4	98.8	94.4
Coronary artery disease	1427	465	67.5	99.4	80.4	94.6	96.2	95.4
Diabetes mellitus	4502	1377	80.6	97.9	88.4	97.0	92.6	94.8
Myocardial infarction	523	282	29.8	86.2	44.2	90.4	76.5	82.9
Chronic kidney disease	640	101	40.8	97.6	57.6	92.9	97.9	95.3
Stroke	693	307	36.5	97.2	53.0	95.7	79.6	87.0
Dementia	317	103	62.1	100.0	76.6	93.1	90.0	91.5
Cataract	240	85	28.6	100.0	44.4	96.1	94.9	95.5
CABG ^b	194	73	32.2	100.0	48.7	96.6	95.0	95.8

^aAll comparisons were significant at $P < .0001$.

^bCoronary artery bypass graft.

Hot off the presses: identification of early GI symptoms via structured vs notes



Courtesy of Jiaxin (Cindy) Gao and Roshni Varma

Hot off the presses: identification of early GI symptoms via structured vs notes

Symptom	Percent of Total Cases Identified as Negative* (n=1,134)	Percent of Total Cases Identified as Positive* (n=1,134)	Proportion of Positive Cases Identified Only by LLM	Proportion of Positive Cases Identified Only by ICD-10	Proportion of Positive Cases Dually Identified
Food Allergies	82.45%	17.55%	0.548 (109/199)	0.040 (8/199)	0.412 (82/199)
Abdominal Distension	87.92%	12.08%	0.482 (66/137)	0.372 (51/137)	0.146 (20/137)
Constipation	41.45%	58.55%	0.708 (354/664)	0.033 (22/664)	0.434 (288/664)
Vomiting	27.60%	72.40%	0.436 (358/821)	0.066 (54/821)	0.498 (409/821)

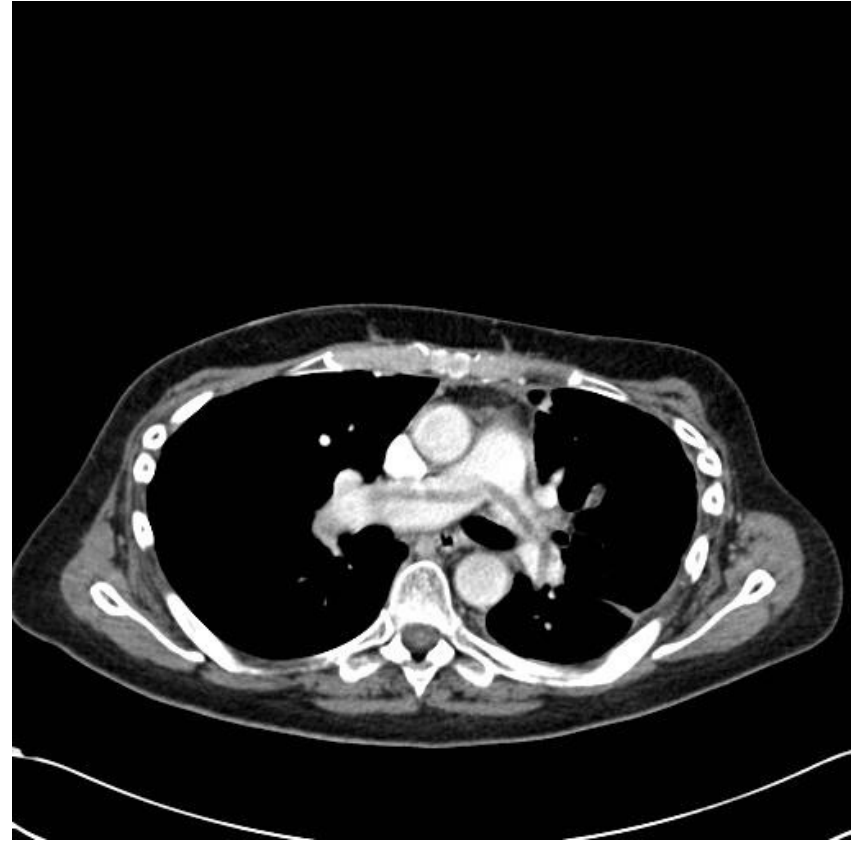
Courtesy of Jiaxin (Cindy) Gao and Roshni Varma

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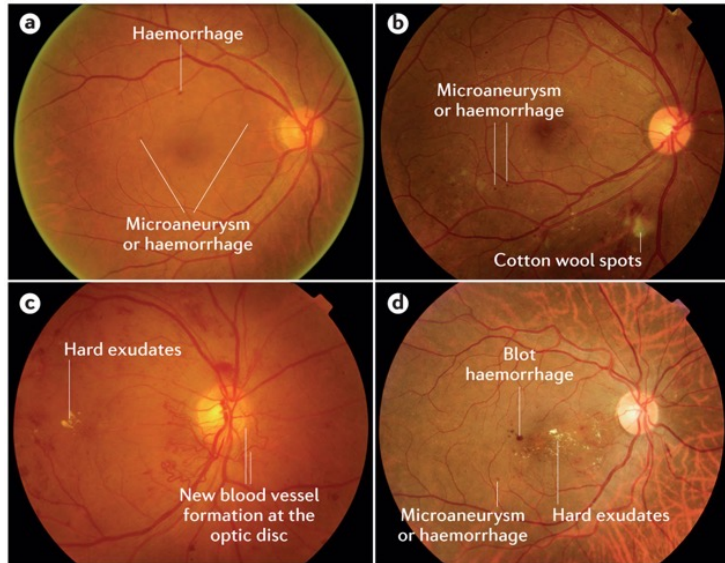
CT / Imaging

- The scans / images themselves
- Interpretation by radiologist (note)



Large non-occlusive pulmonary embolus draped over the bifurcation of the main pulmonary artery; a saddle pulmonary embolus.

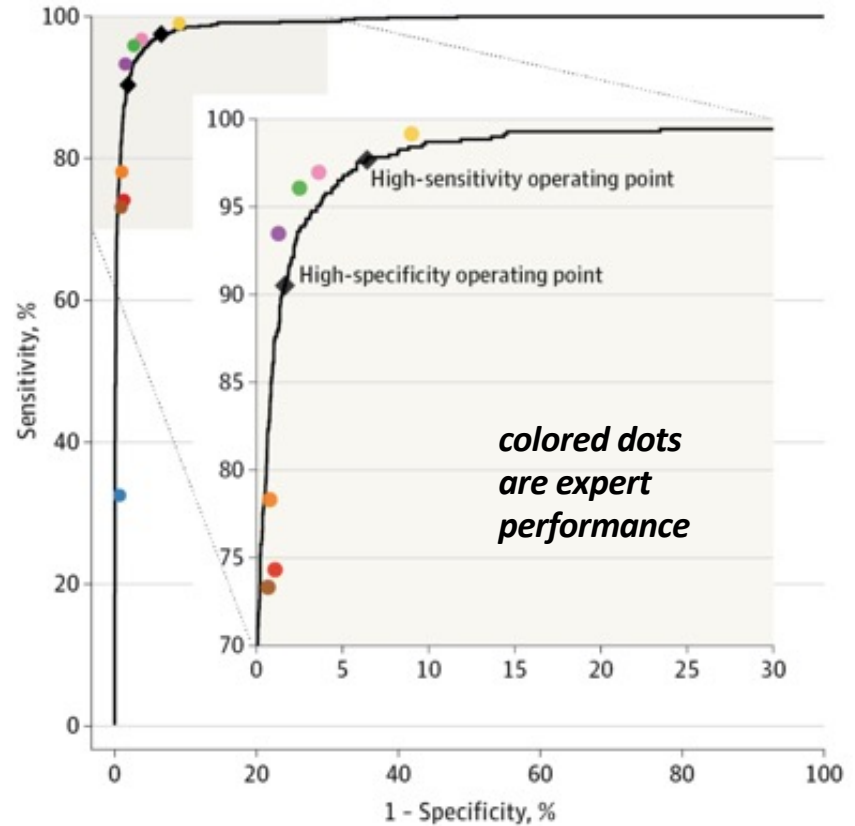
Retinopathy prediction from fundus images



Nature Reviews | Disease Primers

Wong, T. Y. *et al.* (2016) Diabetic retinopathy
Nat. Rev. Dis. Primers doi:10.1038/nrdp.2016.12

A EyePACS-1: AUC, 99.1%; 95% CI, 98.8%-99.3%

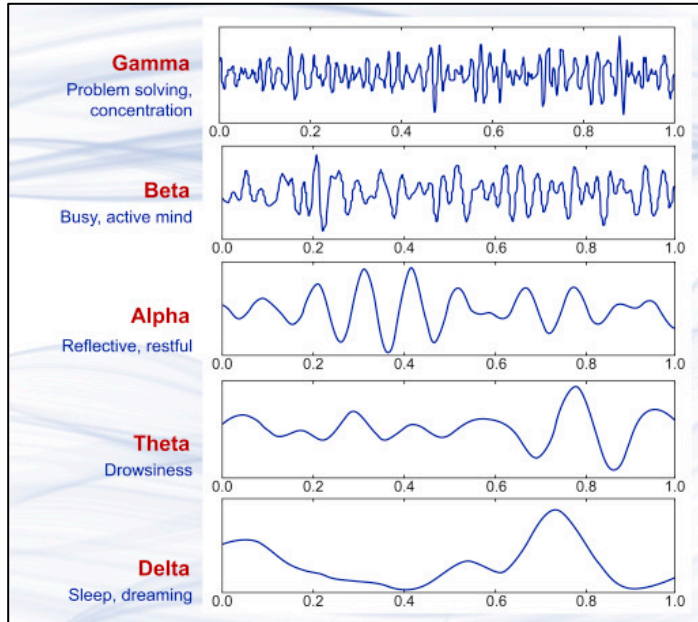


Gulshan V, Peng L, Coram M, et al. Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs. *JAMA*. 2016;316(22):2402-2410. doi:10.1001/jama.2016.17216

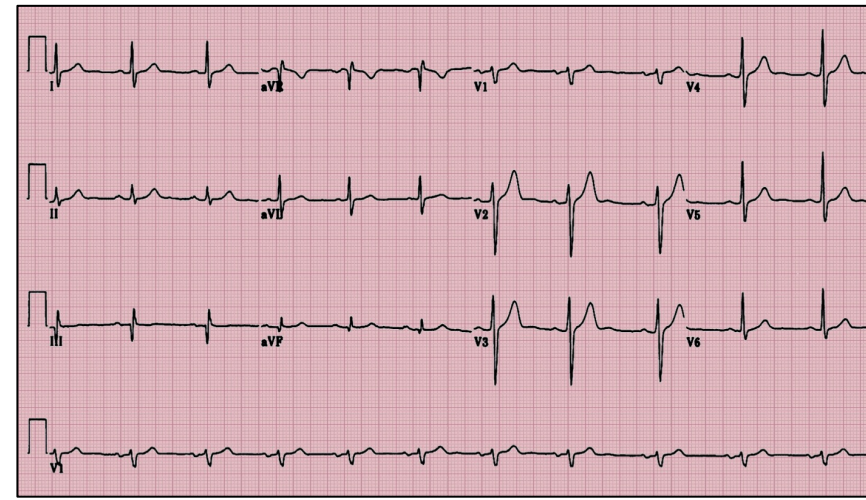
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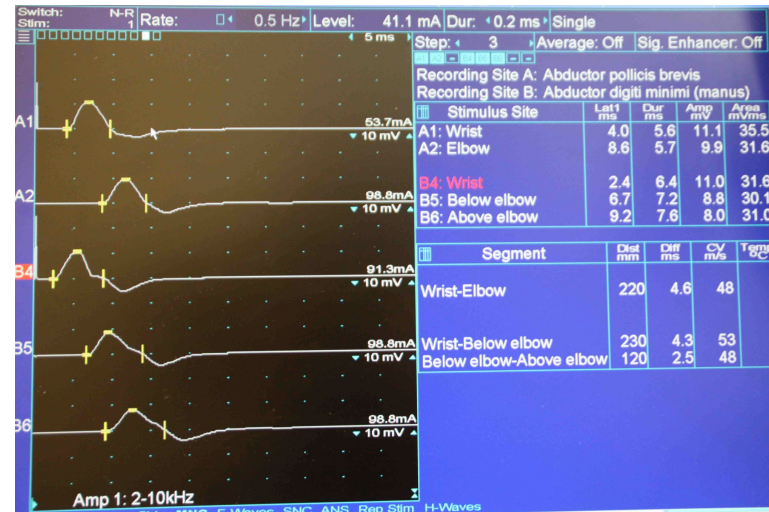
Electrophysiology



EEG



EKG (i.e. ECG)



EMG

Why use unstructured data?

- Reason 1: it contains information not found in structured fields
 - clinical/domain knowledge helps gauge whether/when this info will be useful
 - *example*: use radiological images when predicting COPD prognosis
- Reason 2: it may be available or useful outside of the EHR setting
 - *example*: predict malignancy from images of skin lesions
- Reason 3: it is unaffected (or less affected) by provider decision-making
 - *example*: predict need for surgery from imaging of small bowel obstruction

Part 1: *how*

...do we work with clinical text?

In contrast to structured data...

- Pro: Less manual data wrangling (important caveats apply...)
- Con: Typically larger in size and more difficult to store / load / manage
- Con: Modality-specific knowledge and steps are required to incorporate these data in predictive models

Roadmap: NLP for clinical text

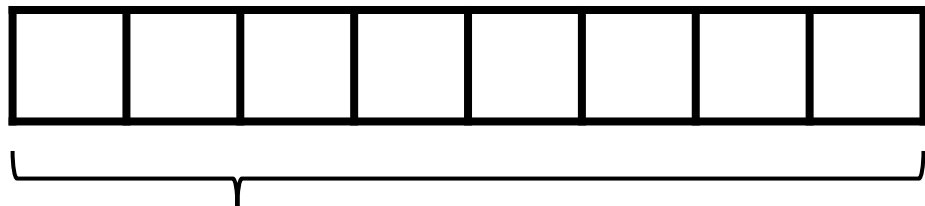
- Today: why use clinical text
- Today: on prediction versus generation
- Today: foundations of NLP and count-based models

- Next time: word embeddings and medical concept embeddings
- After that: current architectures

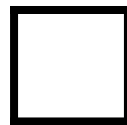
On prediction vs generation

Using current LLMs

Previously: predictive models for tabular data



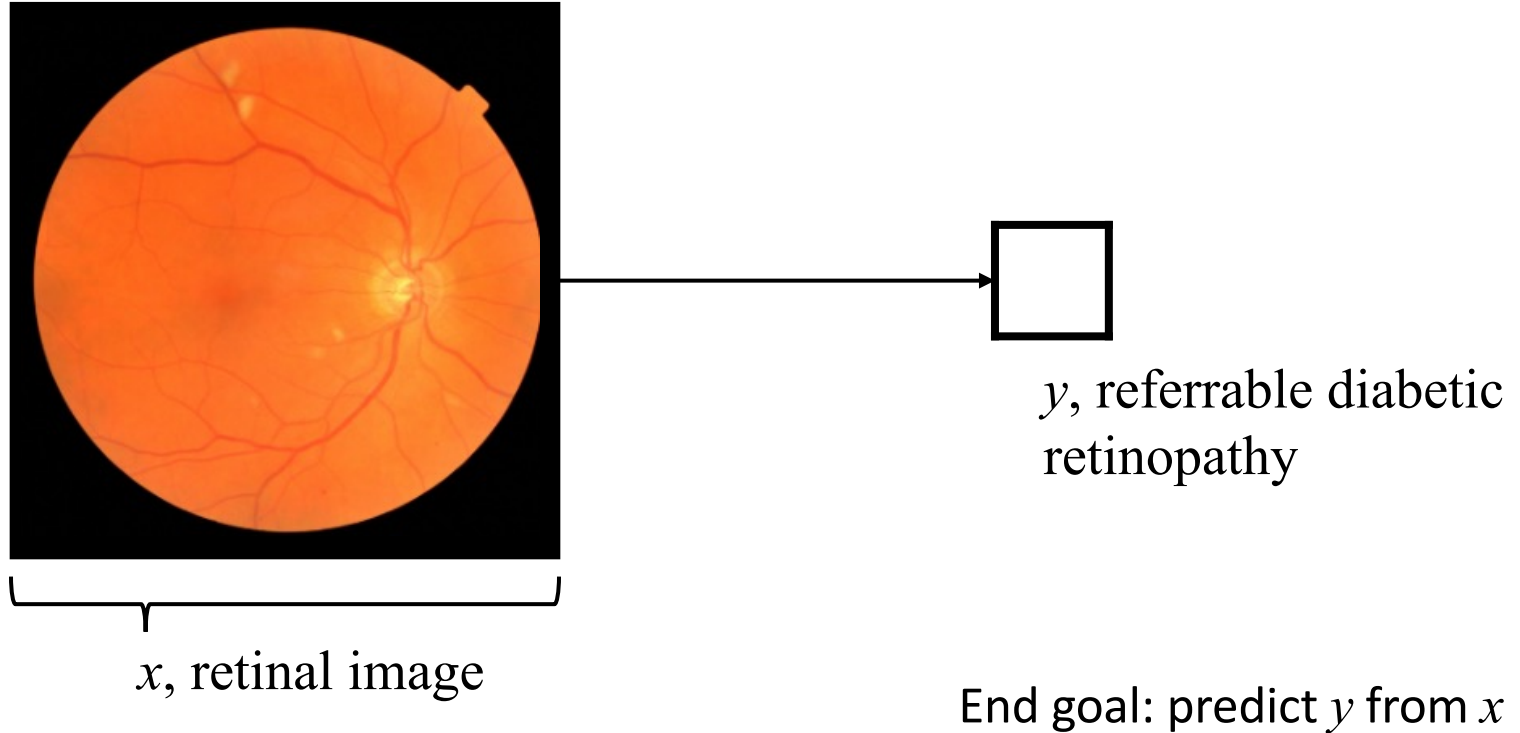
x , data/features for
a subject or patient



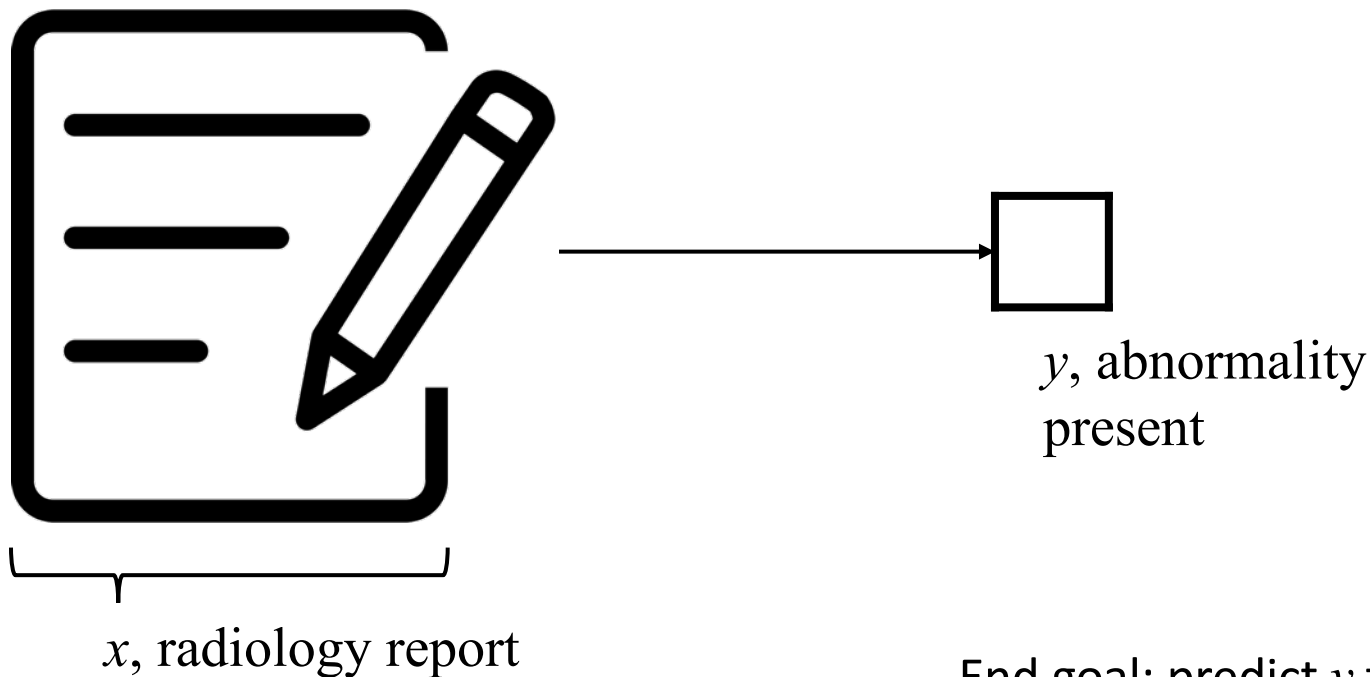
y , associated
value or label

End goal: predict y from x

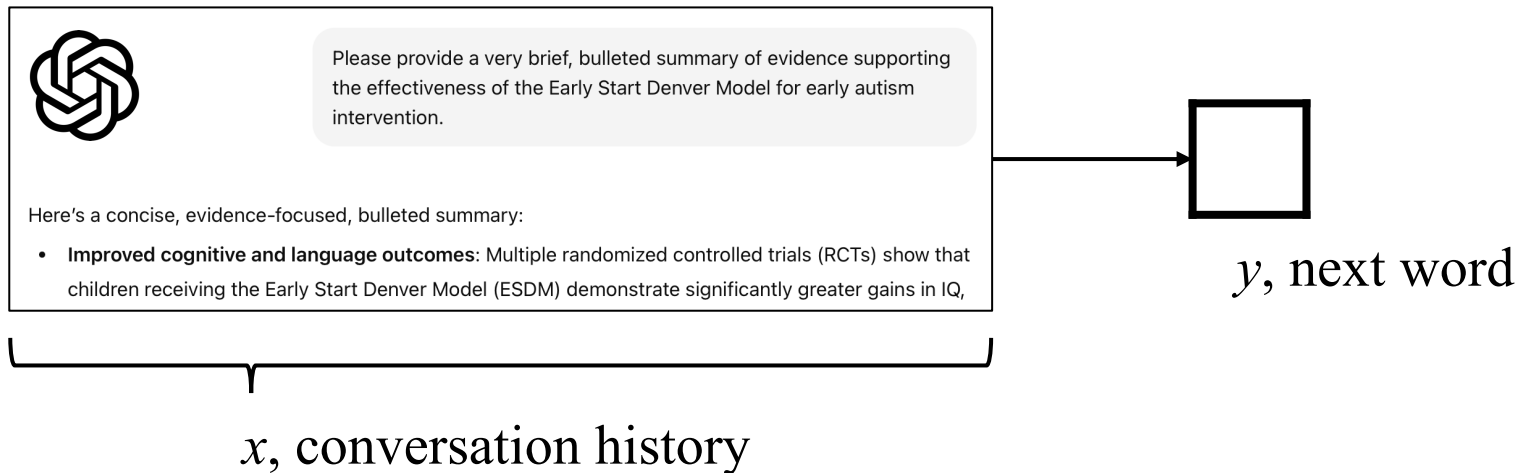
Previously: predictive models for image data



Now: predictive models for text data (NLP)



Is this generative or predictive?



“next-token prediction”

Are all tasks special cases of text generation?

Model: GPT-4

ME

Please read the following echocardiogram report and tell me whether left ventricular hypertrophy was detected. Answer yes or no only. 

Echo reading:

MILD LV SYSTOLIC DYSFUNCTION (See above) WITH MILD LVH
NORMAL RIGHT VENTRICULAR SYSTOLIC FUNCTION MODERATE
VALVULAR REGURGITATION (See above) NO VALVULAR STENOSIS
NO PERICARDIAL EFFUSION SEVERE BIATRIAL ENLARGEMENT AT
LEAST MILD PULMONARY HTN ELEVATED RA PRESSURES NO
PRIOR STUDY FOR COMPARISON



Yes



Are all tasks special cases of text generation?

ME



Please read the following echocardiogram report again and tell me whether left ventricular hypertrophy was detected. This time, please give me your confidence level as a percentage ranging from 0% (i.e. you are certain that it was NOT detected) to 100% (i.e. you are certain that it WAS detected). Answer with the percentage only.

Echo reading:

MILD LV SYSTOLIC DYSFUNCTION (See above) WITH MILD LVH
NORMAL RIGHT VENTRICULAR SYSTOLIC FUNCTION MODERATE
VALVULAR REGURGITATION (See above) NO VALVULAR STENOSIS
NO PERICARDIAL EFFUSION SEVERE BIATRIAL ENLARGEMENT AT
LEAST MILD PULMONARY HTN ELEVATED RA PRESSURES NO
PRIOR STUDY FOR COMPARISON



100%



Are all tasks special cases of text generation?

ME

OK. Now, please read the following text message and tell me whether you believe the sender was angry when sending it. Please give me your confidence level as a percentage ranging from 0% (i.e. you are certain that they were NOT angry) to 100% (i.e. you are certain that they WERE angry). Answer with the percentage only.



> Hey, you didn't show up today. What's the deal?

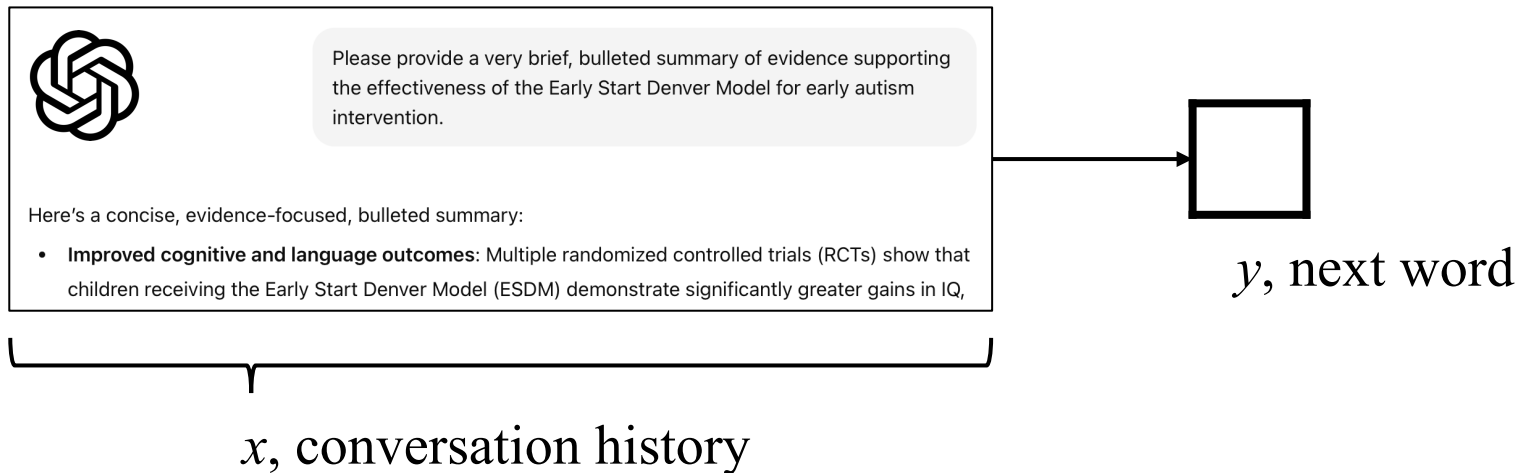


60%



A look ahead:

LLMs are initially trained to predict the next word

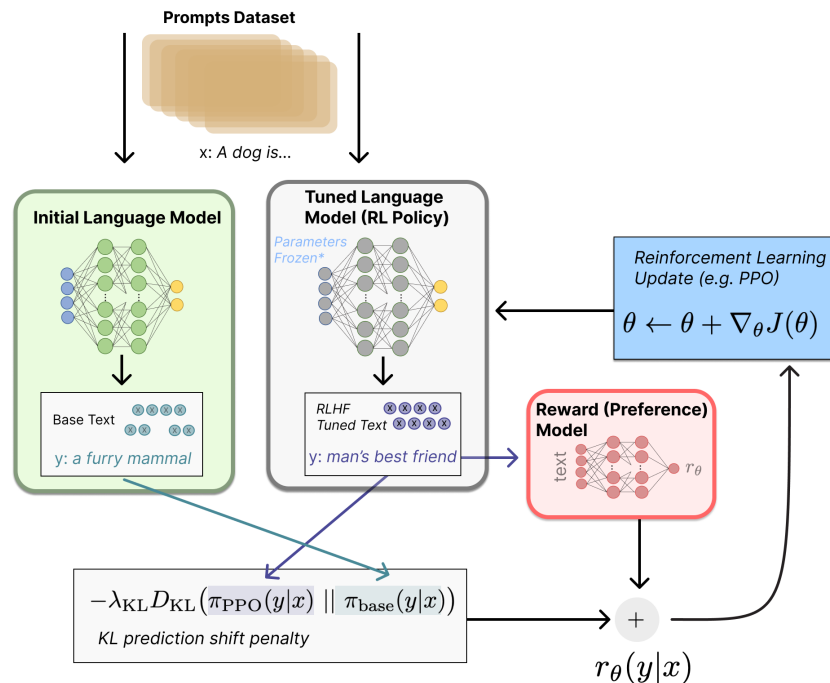


“next-token prediction”

They're then fine-tuned to match our preferences

Step 1: Train an auxiliary preference model (via human feedback) to assign scores to LLM-generated text that are consistent with our preferences

Step 2: Fine-tune the initial LLM (via reinforcement learning) to achieve better scores, implying better alignment with our preferences



Illustrating Reinforcement Learning from Human Feedback (RLHF)
<https://huggingface.co/blog/rlhf>

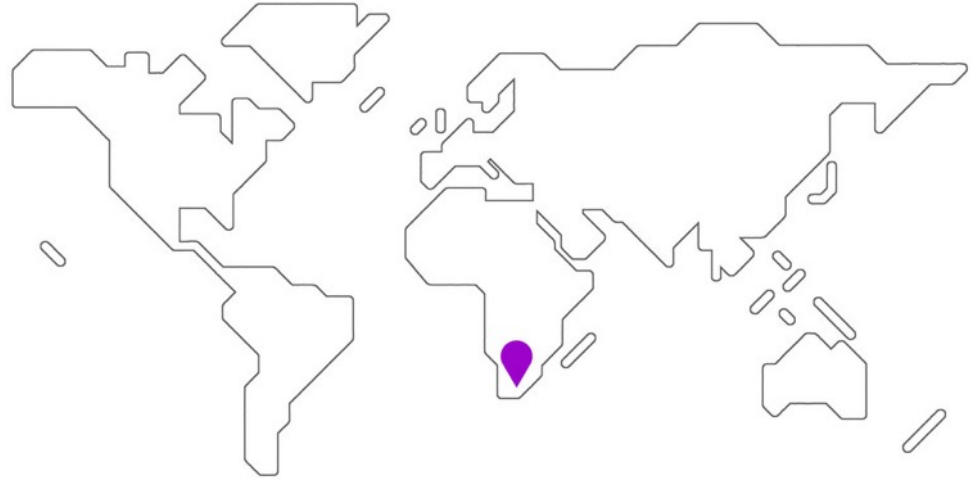
Foundations of NLP

A count-based model for maternal health triage

Case Study: SMS Triage for Global Maternal Health

Maternal Health HelpDesk:

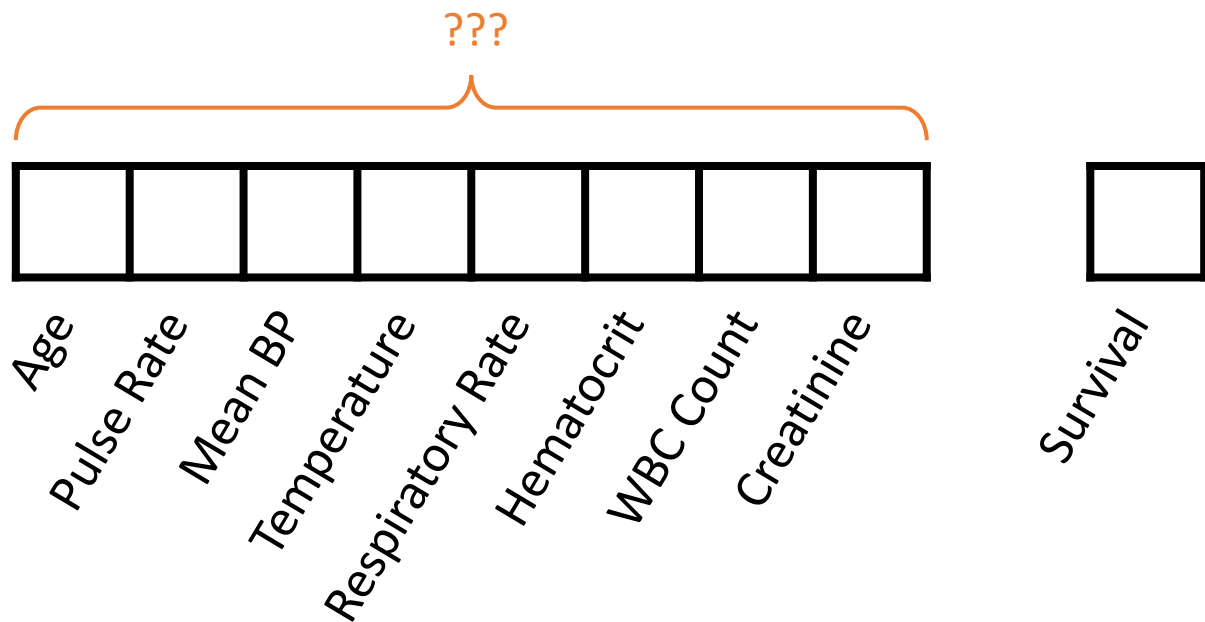
**2 million women connected to
NDoH staff via SMS**



<https://www.praekelt.org>

Binary Classification: Urgent Message? (Yes/No)

A Simple Predictive Model: ICU Mortality



End goal: predict odds of hospital mortality

Training Set (Historical Data)

x_1										y_1
x_2										y_2
x_3										y_3
x_4										y_4
	$\vdots$									$\vdots$
x_{N-1}										y_{N-1}
x_N										y_N

Find an equation that predicts y based on x across the training set

Making Predictions for New x

x_1											y_1
x_2											y_2
x_3											y_3
x_4											y_4
x_{N-1}											y_{N-1}
x_N											y_N

Find an equation that predicts y based on x across the training set

x_{N+1}											y_{N+1}
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<- Learn to predict new y

Case Study: SMS Triage for Global Maternal Health



<https://www.praekelt.org>

Can we use a standard predictive model
setup to solve this problem?

This time, our training data is text

x_1 What helps with morning sickness? ☐ y_1

x_2 How many months should I breastfeed? ☐ y_2

x_3 I passed out and Mom said I was shaking ☐ y_3

x_4 Where is the nearest clinic? ☐ y_4

⋮

⋮

x_{N-1} I am having heavy bleeding, what should I do? ☐ y_{N-1}

x_N What foods should I eat while pregnant? ☐ y_N

y_i : Urgent or
Not Urgent?

x_{N+1} My heart is racing and I can't catch my breath ☐ y_{N+1}

<- Learn to predict new y

We need numbers, not words

- **Can we convert our text to a vector or sequence of numbers?**
- If yes, we can use logistic regression (or any other predictive model)!

First try: count words in each SMS

Step 1: Define a vocabulary of words

x_1

What helps with morning sickness?

x_2

How many months should I breastfeed?

x_3

I passed out and Mom said I was shaking

x_4

Where is the nearest clinic?

list of all words
(in no particular order)

shaking
what
clinic
how
helps
was
nearest
many

with
said
months
the
morning
mom
should
sickness

and
I
is
how
out
breastfeed
passed
where

Step 2: count how many times each vocabulary word appears in a given SMS

What helps with morning sickness?

x_I

0	1	0	0	1	0	0	0	1	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0
shaking	what	clinic	how	helps	was	nearest	many	with	said	months	the	morning	mom	should	sickness	and	I	is	how	out	breastfeed	passed	where

Step 2: count how many times each vocabulary word appears in a given SMS

I passed out and Mom said I was shaking

x_3

1	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	2	0	0	1	0	1	0
shaking	what	clinic	how	helps	was	nearest	many	with	said	months	the	morning	mom	should	sickness	and	I	is	how	out	breastfeed	passed	where

Note that word order does not matter!

clinic is where nearest the

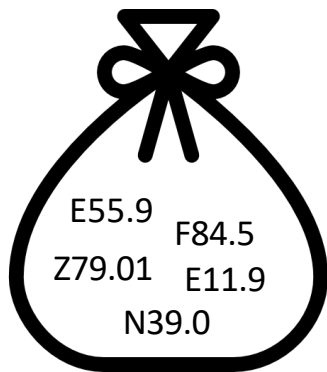
x_4

0	0	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	1
shaking	what	clinic	how	helps	was	nearest	many	with	said	months	the	morning	mom	should	sickness	and	I	is	how	out	breastfeed	passed	where

A “bag of words”



A “bag of medical codes”



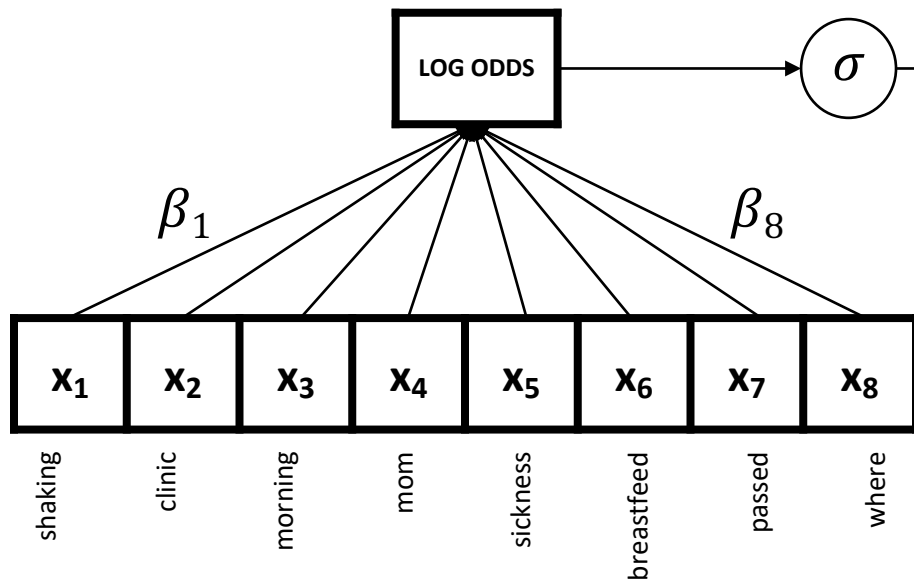
(you did this in coding assignment 1)

PCOR_RSH_DIAGNOSIS_VW EXAMPLE

DIAGNOSISID	PATID	ENCOUNTERID	ENC_TYPE	ADMIT_DATE	PROVIDERID	DX	DX_TYPE	DX_SOURCE	DX_ORIGIN	PDX	DX_POA	RAW_DX	RAW_DX_TYPE	RAW_DX_SOURCE	RAW_PDX
36510180	123456	57896636	AV	2014/02/21 00:00:00	1745326	530.9	9	FI	BI	OT	NI	530.9	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36510407	223456	57897312	AV	2014/05/07 00:00:00	73411	530.81	9	FI	BI	OT	NI	530.81	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36510405	323456	57894371	ED	2014/06/13 00:00:00	123901	553.3	9	FI	BI	OT	NI	553.3	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY
26510406	423456	57843289	AV	2015/01/21 00:00:00	1719263	530.3	9	FI	BI	OT	NI	530.3	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY
36510721	523456	57898224	AV	2015/03/04 00:00:00	1719263	787.2	9	FI	BI	OT	NI	787.2	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36510720	612345	57899120	ED	2015/07/30 00:00:00	156342	454.8	9	FI	BI	OT	NI	454.8	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY
36511049	654321	57892268	EI	2016/12/03 00:00:00	165493	V12.72	9	FI	BI	OT	NI	V12.72	ICD-9-CM	FINAL DIAGNOSIS	SECONDARY
36508682	543216	57893648	IP	2016/08/05 00:00:00	178238	211.3	9	FI	BI	OT	NI	211.3	ICD-9-CM	FINAL DIAGNOSIS	PRIMARY

Now we can use logistic regression.

$$\text{URGENCY LOG ODDS} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots$$



y , associated label:
(0 = not urgent, 1 = urgent)

Variation on counts:

term frequency-inverse document frequency (tf-idf)

- What helps with morning sickness?
- How many months should I breastfeed?
- I passed out and Mom said I was shaking
- Where is the nearest clinic?
- I am having heavy bleeding, what should I do?
- What foods should I eat while pregnant?
- My heart is racing and I can't catch my breath

$\frac{\text{term frequency}}{\text{document frequency}}$ for 'shaking'

$$\frac{1/9}{1/7} = .78$$

$\frac{\text{term frequency}}{\text{document frequency}}$ for 'I'

$$\frac{2/9}{5/7} = .31$$

Count-based feature extraction is still useful!

- Entirely data-driven
 - Vocabulary of words we care about is derived from the training data
 - We then represent text as counts of each vocabulary word
 - We can also count 2- and 3-word phrases; this helps with negation and context
- Knowledge-driven extraction of key words or concepts
 - Rather than creating a vocabulary from the data, we can identify words we (or content experts) believe are important for a given task
 - Concept extraction systems (e.g. cTakes) will identify many alternative phrasings for the same clinical concept (e.g. diagnosis) and group them together as a single feature

Discuss: pros and cons of the bag of words approach

This is important because:

- Understanding the limitations of this approach is important before we consider how current approaches overcome them
- Simple, bag of words models are still appropriate in some cases, and they are highly interpretable

Strengths and Weaknesses

- (+) Count-based approaches are simple and work surprisingly well in practice
- (+) Often the best approach with small datasets
- (-) Does not capture word order or interpret words in context
- (-) Does not group synonyms together or understand semantic relationships between words

Next time:

word and concept embeddings